Convolutional neural networks

3 minutes

The ability to use filters to apply effects to images is useful in image processing tasks, such as you might perform with image editing software. However, the goal of computer vision is often to extract meaning, or at least actionable insights, from images; which requires the creation of machine learning models that are trained to recognize features based on large volumes of existing images.

> **Tip**
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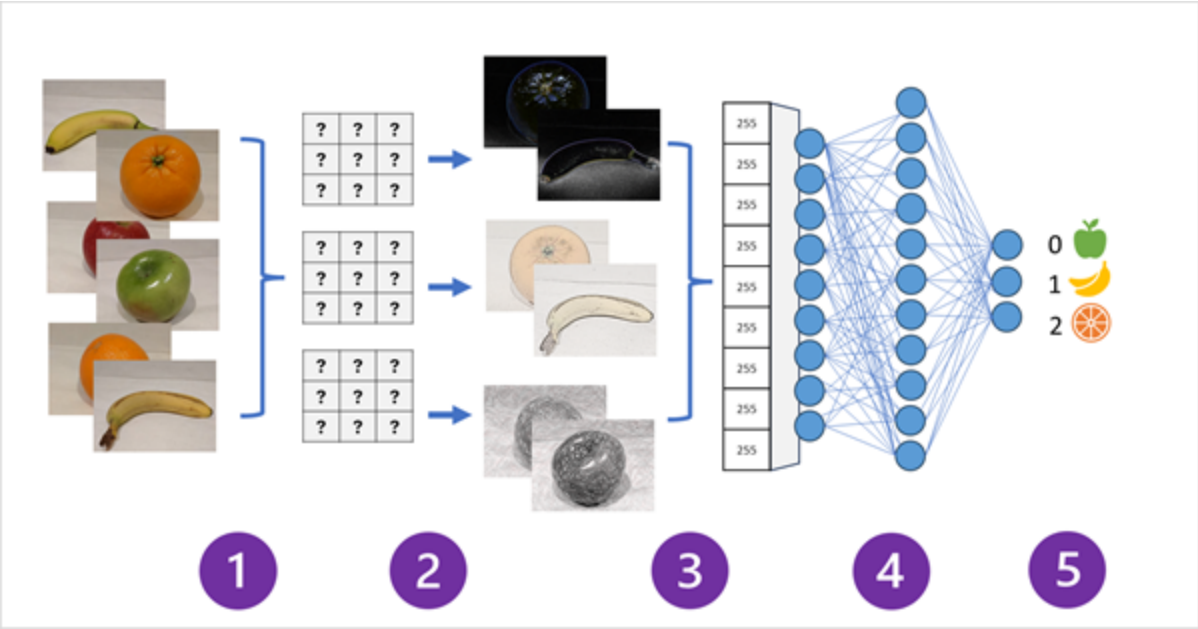
> This unit assumes you are familiar with the fundamental principles of machine learning, and that you have conceptual knowledge of deep learning with neural networks. If you are new to machine learning, consider completing the **Introduction to machine learning concepts** module on Microsoft Learn.

One of the most common machine learning model architectures for computer vision is a *convolutional neural network* (CNN), a type of deep learning architecture. CNNs use filters to extract numeric feature maps from images, and then feed the feature values into a deep learning model to generate a label prediction. For example, in an *image classification* scenario, the label represents the mai subject of the image (in other words, what is this an image of?). You might train a CNN model with images of different kinds of fr (such as apple, banana, and orange) so that the label that is predicted is the type of fruit in a given image.

During the *training* process for a CNN, filter kernels are initially defined using randomly generated weight values. Then, as the training process progresses, the models predictions are evaluated against known label values, and the filter weights are adjusted to improve accuracy. Eventually, the trained fruit image classification model uses the filter weights that best extract features that help identify different kinds of fruit.

The following diagram illustrates how a CNN for an image classification model works:



1. Images with known labels (for example, 0: apple, 1: banana, or 2: orange) are fed into the network to train the model.
2. One or more layers of filters is used to extract features from each image as it is fed through the network. The filter kernels start with randomly assigned weights and generate arrays of numeric values called *feature maps*. Additional layers may "pool" or "downsize" the feature maps to create smaller arrays that emphasize the key visual features extracted by the filters.
3. The feature maps are flattened into a single dimensional array of feature values.
4. The feature values are fed into a fully connected neural network.
5. The output layer of the neural network uses a *softmax* or similar function to produce a result that contains a probability value for each possible class, for example [0.2, 0.5, 0.3].

During training the output probabilities are compared to the actual class label - for example, an image of a banana (class 1) should have the value [0.0, 1.0, 0.0]. The difference between the predicted and actual class scores is used to calculate the *loss* in the model, and the weights in the fully connected neural network and the filter kernels in the feature extraction layers are modified to reduce the loss.

The training process repeats over multiple *epochs* until an optimal set of weights has been learned. Then, the weights are saved and the model can be used to predict labels for new images for which the label is unknown.

> **Note**
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> CNN architectures usually include multiple convolutional filter layers and additional layers to reduce the size of feature maps, constrain the extracted values, and otherwise manipulate the feature values. These layers have been omitted in this simplified example to focus on the key concept, which is that filters are used to extract numeric features from images, which are then used in a neural network to predict image labels.

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